
Problem Set 5

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Problem 5

1.

$$\left(S_x^{(1)} \otimes S_y^{(2)}\right) (|+\rangle_1 \otimes |+\rangle_2) = \left(S_x^{(1)} |+\rangle_1\right) \otimes \left(S_y^{(2)} |+\rangle_2\right) = \frac{\hbar}{2} |-\rangle_1 \otimes \frac{i\hbar}{2} |-\rangle_2.$$

2. The norm squared is

$$\left(\frac{\hbar}{2} \langle - |_1 \otimes \frac{-i\hbar}{2} \langle - |_2\right) \left(\frac{\hbar}{2} |-\rangle_1 \otimes \frac{i\hbar}{2} |-\rangle_2\right) = \frac{\hbar^2}{4} \langle - | - \rangle_1 \cdot \frac{\hbar^2}{4} \langle - | - \rangle_2 = \frac{\hbar^4}{16}.$$

Therefore, the norm is $\frac{\hbar^2}{4}$.

Problem 2

1. Note that $P_+^{(1)} = |+\rangle_1 \langle +|_1$. Moreover, factoring gives $|\psi\rangle = \frac{1}{\sqrt{2}} (|+\rangle_1 + |-\rangle_1) \otimes |+\rangle_2$.

$$\begin{aligned} \mathcal{P}_+ &= \langle \psi | P_+^{(1)} \otimes \mathbb{1}^{(2)} | \psi \rangle \\ &= \langle \psi | \left(P_+^{(1)} | \psi \rangle_1 \otimes \mathbb{1}^{(2)} | \psi \rangle_2 \right) \\ &= \left({}_1 \langle \psi | P_+^{(1)} | \psi \rangle_1 \right) \left({}_2 \langle \psi | \psi \rangle_2 \right) \\ &= {}_1 \langle \psi | + \rangle_1 {}_1 \langle + | \psi \rangle_1 \\ &= |{}_1 \langle \psi | + \rangle_1|^2 \\ &= \left| \left(\frac{1}{\sqrt{2}} {}_1 \langle + | + \rangle_1 + {}_1 \langle - | + \rangle_1 \right) | + \rangle_1 \right|^2 \\ &= \left| \frac{1}{\sqrt{2}} {}_1 \langle + | + \rangle_1 \right|^2 \\ &= \frac{1}{2}. \end{aligned}$$

The normalized state is then $|+\rangle_1 \otimes |+\rangle_2$, since we have collapsed the first electron into $|+\rangle$ by measuring it, and have not measured the second electron at all, thus not changing its state. Since the only other possibility is measuring $-\hbar/2$, and probabilities sum to 1, we have that $\mathcal{P}_- = 1/2$ as well, and by the same logic that normalized state post-measurement would be $|-\rangle_1 \otimes |+\rangle_2$.

2. Note that

$$\frac{1}{2} (|+\rangle_1 + |-\rangle_1) \otimes (|+\rangle_2 + |-\rangle_2) = \frac{1}{\sqrt{2}} (|+\rangle_1 + |-\rangle_1) \otimes \frac{1}{\sqrt{2}} (|+\rangle_2 + |-\rangle_2).$$

This follows from bilinearity of the tensor product and I will use it to justify the fact that ${}_2\langle\psi|\psi\rangle_2 = 1$, since this makes both individual electron states clearly normalized.

$$\begin{aligned}
\mathcal{P}_+ &= \langle\psi| \left(P_+^{(1)} \otimes \mathbb{1}^{(2)} \right) |\psi\rangle \\
&= \langle\psi| \left(P_+^{(1)}|\psi\rangle_1 + \mathbb{1}^{(2)}|\psi\rangle_2 \right) \\
&= {}_1\langle\psi|P_+^{(1)}|\psi\rangle_1 {}_2\langle\psi|\mathbb{1}^{(2)}|\psi\rangle_2 \\
&= {}_1\langle\psi|+\rangle_1 {}_1\langle+|\psi\rangle_1 {}_2\langle\psi|\psi\rangle_2 \\
&= |{}_1\langle+|\psi\rangle_1|^2 \\
&= \left| \frac{1}{\sqrt{2}} {}_1\langle+|\psi\rangle_1 \right|^2 \\
&= \frac{1}{2}.
\end{aligned}$$

Once again, the normalized state post-measurement is simply $|+\rangle_1 \otimes \frac{1}{\sqrt{2}}(|+\rangle_2 + |-\rangle_2)$, as we have collapsed the first electron to $|+\rangle$, and by my earlier restatement of $|\psi\rangle$, the resultant state from doing this is still normalized. Since probabilities sum to 1, $\mathcal{P}_- = \frac{1}{2}$ as well, and the collapsed state after that measurement would be $|-\rangle_1 \otimes \frac{1}{\sqrt{2}}(|+\rangle_2 + |-\rangle_2)$ by the same logic.

3. This one is just going to be messy. I will omit the subscripts for now because I just don't want to type them.

$$\begin{aligned}
\langle\psi|P_+ \otimes \mathbb{1}|\psi\rangle &= \langle\psi| (P_+ \otimes \mathbb{1}) \left(\frac{1}{\sqrt{2}} (|+\rangle \otimes |+\rangle) + (|-\rangle \otimes |-\rangle) \right) \\
&= \frac{1}{\sqrt{2}} \langle\psi| ((P_+ \otimes \mathbb{1}) (|+\rangle \otimes |+\rangle) + (P_+ \otimes \mathbb{1}) (|-\rangle \otimes |-\rangle)) \\
&= \frac{1}{\sqrt{2}} \langle\psi| ((P_+|+\rangle \otimes \mathbb{1}|+\rangle) + (P_+|-\rangle \otimes \mathbb{1}|-\rangle)) \\
&= \frac{1}{\sqrt{2}} \langle\psi| ((|+\rangle \otimes |+\rangle) + (0 \otimes |-\rangle)) \\
&= \frac{1}{2} ((\langle+|\otimes\langle+|) + (\langle-|\otimes\langle-|)) (|+\rangle \otimes |+\rangle) + (0 \otimes |-\rangle)) \\
&= \frac{1}{2} ((\langle+|\otimes\langle+|) (|+\rangle \otimes |+\rangle) + (\langle+|\otimes\langle+|) (0 \otimes |-\rangle) \\
&\quad + (\langle-|\otimes\langle-|) (|+\rangle \otimes |+\rangle) + (\langle-|\otimes\langle-|) (0 \otimes |-\rangle)) \\
&= \frac{1}{2} (|\langle+|\psi\rangle|^2 + \langle+|0\rangle\langle+|-\rangle + |\langle-|\psi\rangle|^2 + \langle-|0\rangle\langle-|+\rangle) \\
&= \frac{1}{2}.
\end{aligned}$$

Once again, since probabilities sum to 1/2, the probability that we measure $-\hbar/2$ in this specific state must also be $\mathcal{P}_- = 1/2$. If we measure $|+\rangle_1$, then the only state in the superposition that we could measure is $|+\rangle_1 \otimes |+\rangle_2$. This can also be seen by noting that in my calculation of $\mathcal{P}_+$, I found that $\left(P_+^{(1)} \otimes \mathbb{1}^{(2)} \right) |\psi\rangle = (|+\rangle \otimes |+\rangle) + (0 \otimes |-\rangle)$. Since the tensor product is bilinear, the product $-\otimes|-\rangle$ must be linear, and linear maps send $0 \mapsto 0$. Therefore, plugging in the zero vector gives $0 \otimes |-\rangle = 0$. We have thus found that measuring $\hbar/2$ in the first

electron, given by the projection operator $P_+^{(1)}$, maps

$$|\psi\rangle \xrightarrow{P_+} |+\rangle_1 \otimes |+\rangle_2$$

Similar logic and calculations show that measuring $-\hbar/2$ yields a collapsed normalized state of $|-\rangle_1 \otimes |-\rangle_2$. I will omit the calculation since it is basically the same and needlessly messy.

4. The calculations here are all basically the same as before, so I will start with the table, and will do the calculations afterwards. I will show less work because once again, they are basically the same as before.

	$\mathcal{P}\left(S_z^{(1)} = \frac{\hbar}{2}, S_z^{(2)} = \frac{\hbar}{2}\right)$	$\mathcal{P}\left(S_z^{(1)} = -\frac{\hbar}{2}, S_z^{(2)} = \frac{\hbar}{2}\right)$	$\mathcal{P}\left(S_z^{(1)} = \frac{\hbar}{2}, S_z^{(2)} = -\frac{\hbar}{2}\right)$	$\mathcal{P}\left(S_z^{(1)} = -\frac{\hbar}{2}, S_z^{(2)} = -\frac{\hbar}{2}\right)$
i	1/2	1/2	0	0
ii	1/4	1/4	1/4	1/4
ii	1/2	0	0	1/2

- (a) Recall the first normalized state after measurement was $|\psi\rangle = |\pm\rangle_1 \otimes |+\rangle_2$. Measuring again in $S_z^{(2)}$ gives a probability of seeing $\hbar/2$ as

$$\begin{aligned} \mathcal{P}_{\pm,+} &= \langle\psi| \left(\mathbb{1}^{(1)} \otimes |+\rangle\langle+|^{(2)} \right) |\psi\rangle \\ &= |{}_2\langle\psi|+\rangle_2|^2 \\ &= |{}_2\langle+|+\rangle_2|^2 \\ &= 1. \end{aligned}$$

Multiplying this by the probability of already having measured a state $|+\rangle_1$ or $|-\rangle_1$ gives probabilities $\mathcal{P}_{+,+} = \mathcal{P}_{-,+} = 1/2$. We also find that the probability of measuring $|-\rangle$ is 0, since probabilities sum to 1.

- (b) The normalized post-measurement states were $|\pm\rangle_1 \otimes \frac{1}{\sqrt{2}}(|+\rangle_2 + |-\rangle_2) = |\pm\rangle \otimes |+_x\rangle_2$. Measuring again in $S_z^{(2)}$ gives

$$\begin{aligned} \mathcal{P}_{\pm,+} &= \langle\psi| \left(\mathbb{1}^{(1)} \otimes |+\rangle\langle+|^{(2)} \right) |\psi\rangle \\ &= |{}_2\langle+_x|+\rangle_2|^2 \\ &= \frac{1}{2}. \end{aligned}$$

Multiplying through by the probability of measuring either $|+\rangle_1$ or $|-\rangle_1$ gives probabilities of $\mathcal{P}_{+,+} = \mathcal{P}_{-,+} = 1/4$. Since probabilities sum to 1, we have $\mathcal{P}_{\pm,-} = 1/2$, and by the same logic we have all other probabilities are 1/4 as well.

- (c) The state post measurement is either $|+\rangle \otimes |+\rangle$ or $|-\rangle \otimes |-\rangle$. Note that

$$\begin{aligned} \mathcal{P}_{++} &= \langle\psi| \left(\mathbb{1}^{(1)} \otimes |+\rangle\langle+|^{(2)} \right) |\psi\rangle \\ &= |{}_2\langle+|+\rangle_2|^2 \\ &= 1. \end{aligned}$$

Multiplying through by the probability of finding $|+\rangle_1 \otimes |+\rangle_2$ in the first place gives a probability of $1/2$. Similarly,

$$\begin{aligned}\mathcal{P}_{--} &= \langle \psi | \left(\mathbb{1}^{(1)} \otimes |-\rangle\langle -|^{(2)} \right) | \psi \rangle \\ &= |2\langle - | - \rangle_2|^2 \\ &= 1.\end{aligned}$$

By the same logic, the true probability here is $1/2$. Since probabilities sum to 0, we have $\mathcal{P}_{+-} = \mathcal{P}_{-+} = 0$.